

Technology Stack

Date	29 September 2026
Team ID	SWUID2026318872
Project Name	A CRM Application for the Chai Shop & Co
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, React.js, Tailwind CSS	HTML5 creates the website structure, CSS3 and Tailwind CSS provide responsive styling, and React.js supports reusable components such as the navigation bar, product cards, menu categories, cart, checkout form, and customer reviews.

2	Backend / Server-Side	<i>Node.js, Express.js, and TypeScript*</i>	<i>Node.js and Express.js will handle user registration, product management, cart operations, orders, payments, delivery details, and contact-form submissions. TypeScript improves code reliability and maintainability</i>
3	Database / Data Storage	<i>PostgreSQL</i>	<i>PostgreSQL will store structured information such as customer accounts, tea products, product categories, prices, stock quantities, orders, payment status, delivery addresses, and reviews. Its relational structure is suitable for connecting customers with their orders and products.</i>
4	Cloud / Hosting / Deployment	<i>Vercel for the frontend and Render for the backend and PostgreSQL database</i>	<i>Vercel provides fast and simple deployment for the frontend. Render can host the backend API and database. This combination is suitable for a student project because it is easy to configure, cost-effective, and can support future growth.</i>
5	Version Control & CI/CD	<i>Git, GitHub, and GitHub Actions</i>	<i>Git and GitHub will maintain the project source code, track changes, and support collaboration. GitHub Actions can automatically test the</i>

			<i>application and check whether the project builds correctly before deployment.</i>
6	Third-Party APIs / Other Tools	<i>Razorpay, Cloudinary, Nodemailer, and Google Maps API</i>	<i>Razorpay will enable online payments in Indian rupees. Cloudinary will store product and food images. Nodemailer will send order confirmations and contact-form emails. Google Maps API can support delivery-location selection or store-location display.</i>